

Lecture 8: The Product Rule

GOAL: Be able to differentiate functions of the form $f(x)g(x)$.

Summary:

$$x \cdot \sin x$$

$$\frac{d}{dx}[f(x)g(x)] = \frac{df}{dx}g(x) + f(x)\frac{dg}{dx}$$

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$$

Think about why this is true, or find f and g where it is false

Question: Does $[f(x)g(x)]' = f'(x)g'(x)$?

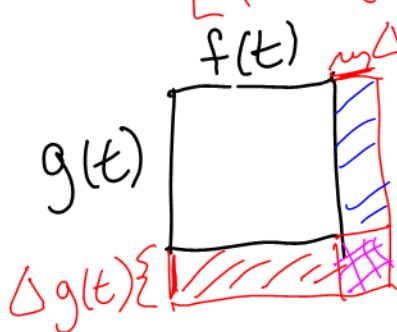
A: No

$$f(x) = 1$$

$$g(x) = x$$

$$[f(x)g(x)]' = [x]' = 1$$

$$[1]' \cdot [x]' = 0 \cdot 1 = 0$$



$\Delta(f(t)g(t)) = \text{Area of the Shaded Region}$

$$= f(t)\Delta g(t) + \Delta f(t)g(t) + \left[\begin{matrix} \rightarrow 0 \\ \Delta t \rightarrow 0 \end{matrix} \right]$$

$$\frac{\Delta(f(t)g(t))}{\Delta t} = f(t)\frac{\Delta g(t)}{\Delta t} + \frac{\Delta f(t)}{\Delta t}g(t) + \left[\text{---} \right]$$

Take $\Delta t \rightarrow 0$

Theorem (Product Rule) Let f and g are differentiable

funcs. Then $\frac{d}{dx}(f(x)g(x)) = \frac{df}{dx}g(x) + f(x)\frac{dg}{dx}$

$$[f(x)g(x)]' = \underbrace{f'(x)g(x)}_{\text{"left" "right"}} + \underbrace{f(x)g'(x)}_{\text{"right d-left plus left d-right"}}$$

Example 1 $\frac{d}{dx}(x^3) = 3x^2$

$$\begin{aligned}\frac{d}{dx}(x^3) &= \frac{d}{dx}(\underbrace{x}_f \cdot \underbrace{x^2}_g) = [x]' \cdot x^2 + x \cdot [x^2]' \\ &= 1 \cdot x^2 + x \cdot (2x) = x^2 + 2x^2 = 3x^2\end{aligned}$$

NOTE: By continuing Ex 1, $\frac{d}{dx}(x^n) = nx^{n-1}$ for an integer $n \geq 0$.

$$\begin{aligned}x^4 &= x \cdot x^3 \\ x^5 &= x \cdot x^4 \\ &\vdots\end{aligned}$$

Example 2 Find y' if $y = \overbrace{2 \sin x}^f \overbrace{\cos x}^g$

$$\begin{aligned}y' &= [2 \sin x]' \cos x + 2 \sin x [\cos x]' \\ &= [2 \cos x] \cos x + 2 \sin x [-\sin x] \\ &= 2 [\cos^2 x - \sin^2 x]\end{aligned}$$

$$y = \underbrace{2}_f \underbrace{\sin x \cos x}_g$$

$$y' = [2]' \sin x \cos x + 2 [\sin x \cos x]'$$

Example 3 Let $h(x) = \underbrace{(x^{10} + x + 1)}_f \cdot \underbrace{(x^2 + 9)}_g$

@ $h'(x)$

$$f(x) = x^{10} + x + 1 \longrightarrow f'(x) = 10x^9 + 1$$

$$g(x) = x^2 + 9 \longrightarrow g'(x) = 2x$$

$$h'(x) = f'(x)g(x) + f(x)g'(x)$$

$$= (10x^9 + 1)(x^2 + 9) + (x^{10} + x + 1)(2x)$$

⑤ Compute $h'(1)$

$$h'(1) = 11 \cdot 10 + 3 \cdot 2 = 116$$

Example 4 Find $h'(x)$ if $h(x) = \underbrace{(x+1)}_f \cdot \underbrace{(2x+1)(3x+1)}_g$

$$\begin{aligned} h'(x) &= (2x+1)(3x+1) + (x+1) \left[\underbrace{(2x+1)}_f \cdot \underbrace{(3x+1)}_g \right]' \\ &= (2x+1)(3x+1) + (x+1) \left[2(3x+1) + (2x+1) \cdot 3 \right] \\ &= 1 \cdot (2x+1)(3x+1) + (x+1) \cdot 2 \cdot (3x+1) + (x+1)(2x+1) \cdot 3 \end{aligned}$$

Example 5 Find the equation for the tangent line to $h(x) = (\sqrt{x} + 4)(x^{\frac{3}{4}} - 2x)$ at $x = 16$

Need $h'(16) = f'(16)g(16) + f(16)g'(16)$

$$f(x) = \sqrt{x} + 4$$

$$f(16) = \sqrt{16} + 4 = 4 + 4 = 8$$

$$f'(x) = [x^{\frac{1}{2}}]' = \frac{1}{2} x^{-\frac{1}{2}}$$

$$= \frac{1}{2\sqrt{x}}$$

$$f'(16) = \frac{1}{2\sqrt{16}} = \frac{1}{2 \cdot 4} = \frac{1}{8}$$

$$g(x) = x^{\frac{3}{4}} - 2x$$

$$= \underbrace{4\sqrt{x}}_{(4\sqrt{x})^3} - 2x$$

$$g(16) = (4\sqrt{16})^3 - 2(16)$$

$$= 8 - 32 = -24$$

$$g'(x) = \frac{3}{4} x^{-\frac{1}{4}} - 2$$

$$= \frac{3}{4 \cdot 4\sqrt{x}} - 2$$

$$g'(16) = \frac{3}{8} - 2 = -\frac{13}{8}$$

$$h'(16) = f'(16)g(16) + f(16)g'(16) = \frac{1}{8} \cdot (-24) + 8 \left(-\frac{13}{8}\right) = -3 - 13 = -16$$

Point: x -coord: 16

$$\begin{aligned} y\text{-coord: } h(16) &= f(16)g(16) \\ &= 8(-24) = -192 \end{aligned}$$

Equation:

$$y - h(16) = h'(16)(x - 16)$$

$$y + 192 = -16(x - 16)$$

$$y + 192 = -16x + 256$$

$$y = -16x + 64$$

Example 6 When does $y = 10x^5 e^x$ have a horizontal tangent line?

$$y' = [10x^5]'e^x + 10x^5 [e^x]'$$

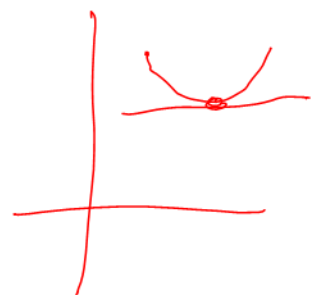
$$= 50x^4 e^x + 10x^5 e^x$$

$$50x^4 e^x + 10x^5 e^x \stackrel{\text{set}}{=} 0$$

$$e^x [50x^4 + 10x^5] = 0$$

$$10x^4 e^x [5 + x] = 0$$

$10 = 0$ ← Never
OR $x^4 = 0$ $x = 0$
OR $e^x = 0$ Never
OR $5 + x = 0$ $x = -5$



Example 7 Compute $h'(x)$ if

$$h(x) = 6e^x \sin x - 13e^x \cos x$$
$$= e^x [6 \sin x - 13 \cos x]$$

$$h'(x) = [e^x]' [6 \sin x - 13 \cos x] + e^x [6 \sin x - 13 \cos x]'$$
$$= e^x [6 \sin x - 13 \cos x] + e^x [6 \cos x + 13 \sin x]$$
$$= e^x [6 \sin x - 13 \cos x + 6 \cos x + 13 \sin x]$$
$$= e^x [19 \sin x - 7 \cos x]$$

$$\left[\frac{f(x)}{g(x)} \right]' = \left[[f(x)] \cdot \left[\frac{1}{g(x)} \right] \right]'$$